

Behavioral Delta Risk

- Calibrated Attrition Scoring for Targeted Retention - Stakeholder Brief

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https://github.com/geobruder/Zeta_WG_Repo
(No tracking - No redirects)

Background

This project evaluates whether short-term behavioral deltas improve attrition prediction beyond static demographic and credit features. The practical goal is a calibrated probability score suitable for a pilot retention campaign. This should serve as a valuable tool for retention management stakeholders who need a reliable prioritized list and probability thresholds to set outreach.

Motivation

- Outreach is costly; prioritization must maximize ROI.
- Short-term behavioral deltas (recent drops in spend/count, fewer contacts) may signal imminent attrition.
- Goal: produce calibrated probabilities for a pilot retention campaign.

Question, Hypothesis, Prediction

- Question: Do short-term behavioral deltas improve attrition prediction beyond static features?
- Hypothesis: Negative deltas increase attrition risk after controlling for demographics/credit.
- Prediction: Models with delta features → higher AUC and lower Brier.

Data at a glance

- Dataset: anonymized BankChurners - (available at source/ in repo)
- Rows in test set: 3039
- Predictor count (approx): see pipeline
- Key fields: demographics, credit metrics, transaction aggregates, quarter-to-quarter deltas.
- Response: target_attrit (1 = attrited).
- Note: PseudoID preserved; identifiers and flagged leakage columns removed.

Engineered features

- Amt_Chg_Q4Q1 — **relative change in transaction amount (Q4 vs Q1).**
- Ct_Chg_Q4Q1 — **relative change in transaction count.**
- Large_Amt_Drop / Large_Ct_Drop — **binary flags (threshold 0.7).**
- Recent_Contact_Reduction — **contacts ≤ 1 .**

Preprocessing & leakage controls

- Median imputation; rare-level consolidation; dummy encoding; zero-variance removal; winsorize outliers.
- Targeted leakage audit removed model-derived scores and identifiers (e.g., Naive_Bayes outputs, CLIENTNUM).
- Transformations implemented in a reproducible recipe applied only on training data.
- - **Final leakage re-audit required** — audit performed; we recommend one more temporal leakage check before production.

Modeling overview

- Baselines: static logistic. Enhanced: logistic + deltas.
- Ensembles: Random Forest (tuned), XGBoost (direct xgboost CV fallback used).
- Validation: stratified 70/30 train/test; repeated 5-fold CV $\times$ 5 for tuning.
- Metrics: AUC (discrimination), Brier (calibration), decision thresholds.
- Implementation detail for reproducibility:
Reconcile caret xgbTree behavior across environments — implemented direct xgboost CV fallback and documented divergence.

Key numeric results (holdout)

Selected model headline metrics:

XGBoost AUC: 0.9927

XGBoost Brier: 0.02282

Summary (from artifacts):

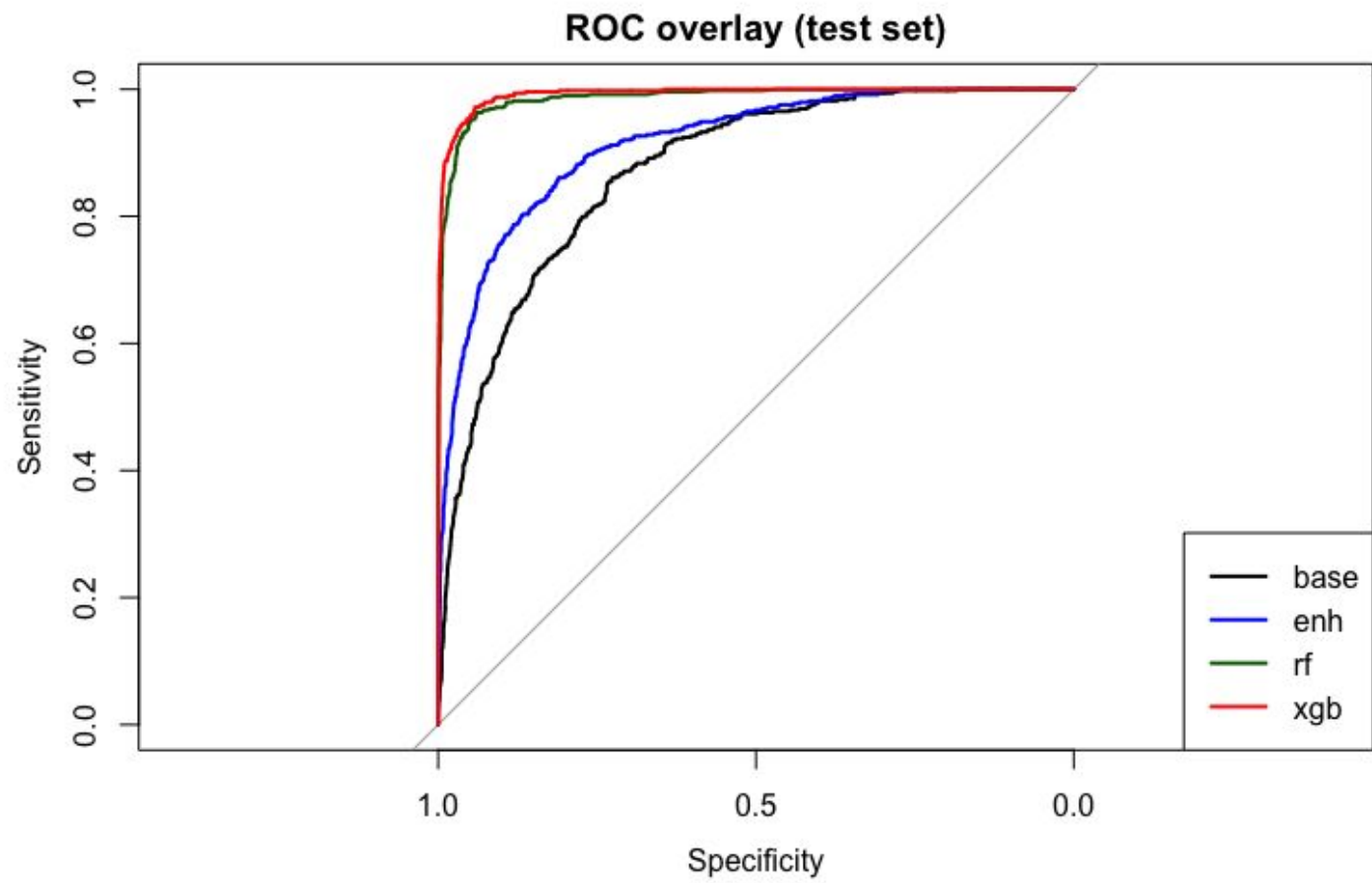
Logit Baseline: **AUC 0.8724 | Brier 0.0941**

Logit Enhanced: **AUC 0.9148 | Brier 0.0739**

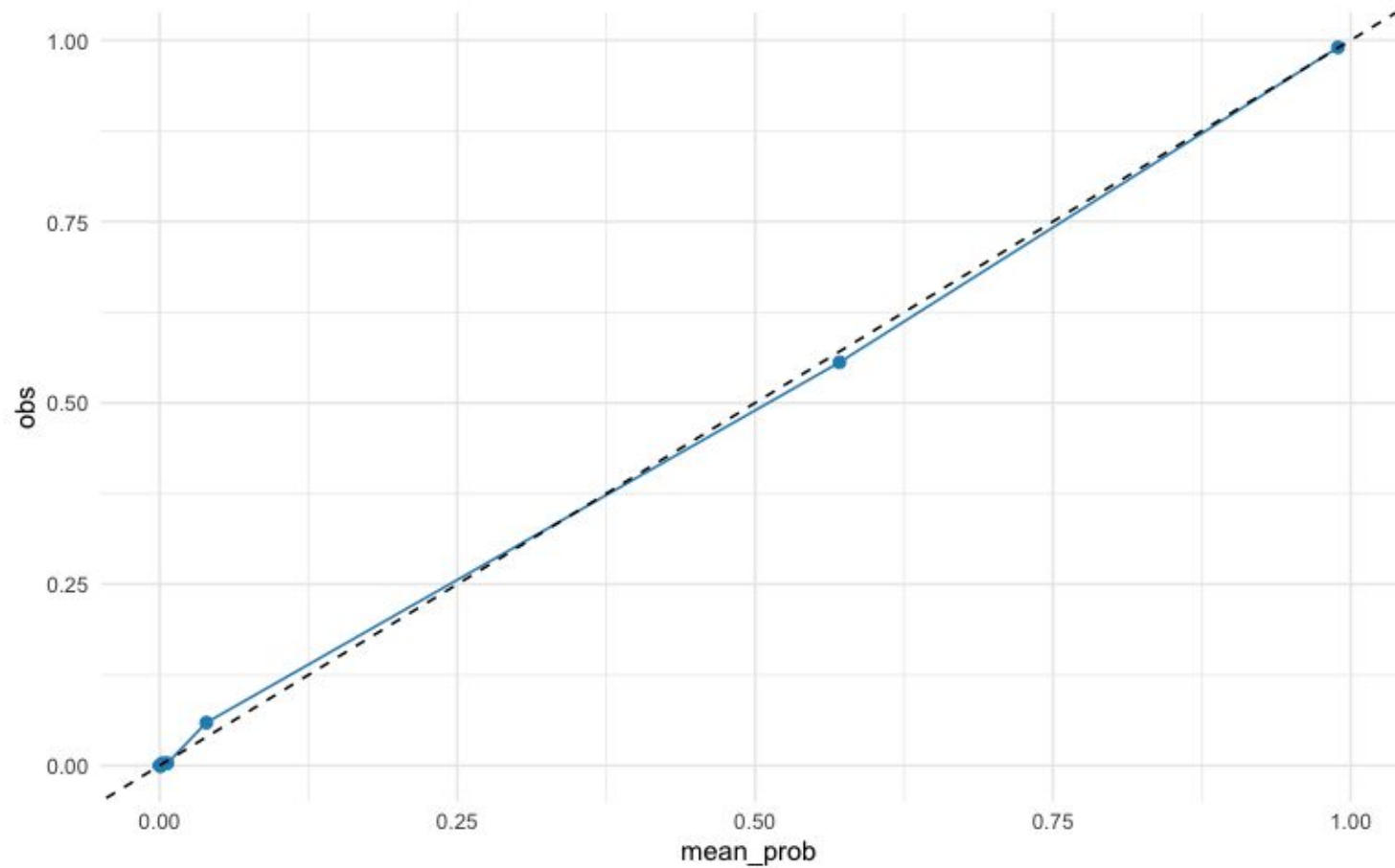
Random Forest (tuned): **AUC $\approx$ 0.9867 | Brier $\approx$ 0.03417**

XGBoost (final, calibrated): **AUC $\approx$ 0.9927 | Brier $\approx$ 0.02282**

Visual diagnostics



Calibration: Reliability diagram



Decision thresholds

(Test set)

threshold	TP	FP	TN	FN	Precision	Recall
0.01	487	458	2092	2	0.5153	0.9959
0.02	486	338	2212	3	0.5898	0.9939
0.05	479	213	2337	10	0.6922	0.9796
0.10	472	143	2407	17	0.7675	0.9652
0.20	462	96	2454	27	0.8280	0.9448

Pilot proposal:

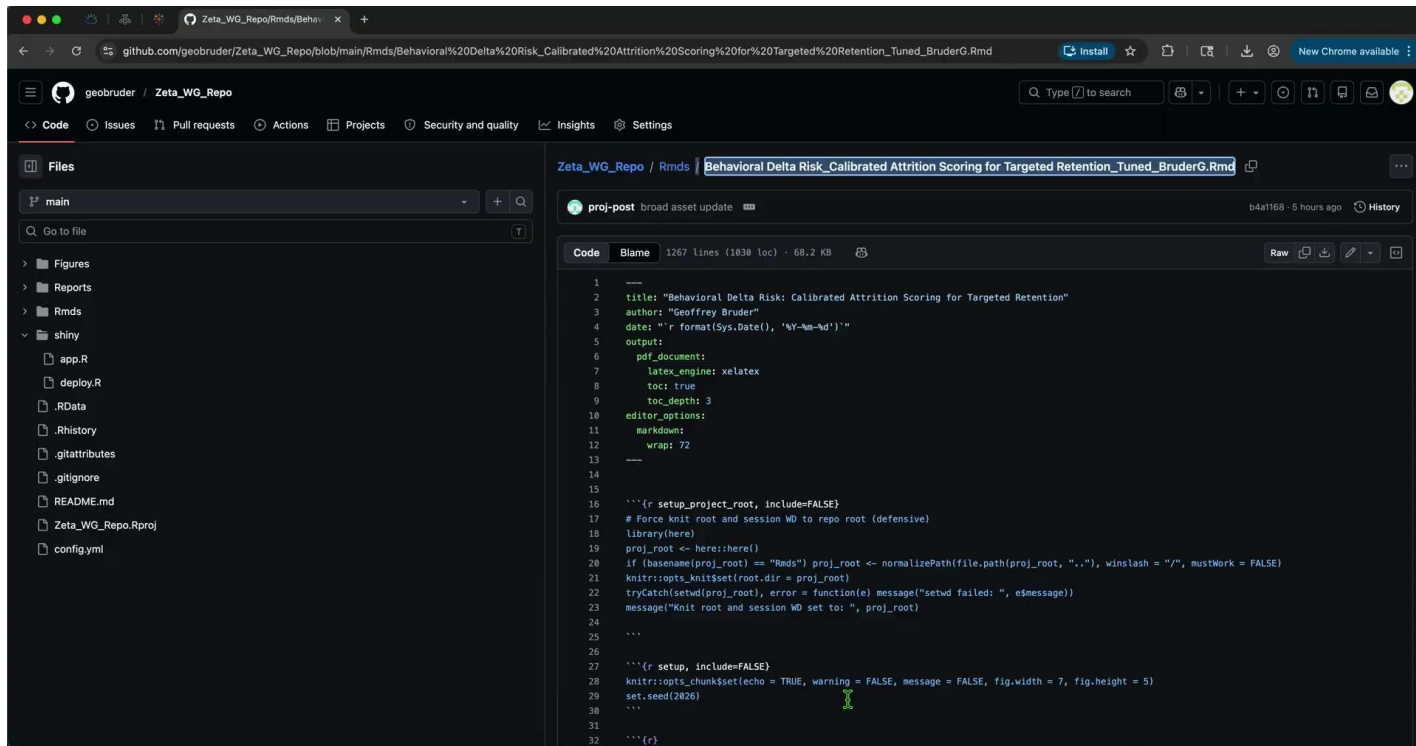
Stage a conservative pilot using calibrated XGBoost at 0.10 threshold;

run A/B test measuring uplift vs control using operational LTV and outreach cost.

App Demo

Live demo will show:

1. Reload artifacts → confirm provenance and `n_test`.
2. Threshold slider → set to **0.10**; show targeted list and ROI preview.
3. Decile filter → inspect top decile and drill to single customer.
4. Model insights → top predictors and local SHAP explanation.



Limitations & cautions

- Exceptionally high ensemble AUCs require final leakage re-audit and temporal validation.
- caret xgbTree inconsistency across environments — reconcile for reproducibility.
- Calibration caution at highest decile — use conservative cutoffs for costly outreach.

Recommendations & next steps

1. Final leakage re-audit and rolling origin validation.
2. Small pilot A/B test using calibrated probabilities and operational LTV/cost inputs. See pilot plan in repo.
3. Reconcile tuning across environments; document final hyper-parameters.
4. Deploy lightweight Shiny app for ops + publish Tableau dashboard; implement monitoring (weekly AUC, decile drift).

Appendix

Clone the repo at [<https://github.com/geobruder/Zeta_WG_Repo>](https://github.com/geobruder/Zeta_WG_Repo), ensure `artifacts/project_artifacts.RData` is in `artifacts/`, run the Shiny app (`shiny::runApp("shiny_app")` or open `app.R`), and explore the Threshold slider (set to 0.10), Decile filter (top decile), ROI preview, and customer drilldown with local explanations.

App includes built-in A/B test scaffolding and monitoring hooks (artifacted model versions, timestamped runs, and weekly AUC/decile drift metrics) to support a staged pilot and ongoing performance tracking.

Tableau Dashboard (UC) is also available for future exploratory analysis and dashboarding of decile performance, cohort trends, and ROI scenarios; use it to supplement the live demo once finalized.

(https://public.tableau.com/app/profile/geoffrey.bruder/viz/BehavioralDeltaRisk_CalibratedAttritionScoringforTargetedRetention_GB/Dashboard1)



Shiny App Code
(@ Project Repo)



Project Repository
(view README)



Future Tableau Dashboard
(UNDER CONSTRUCTION)